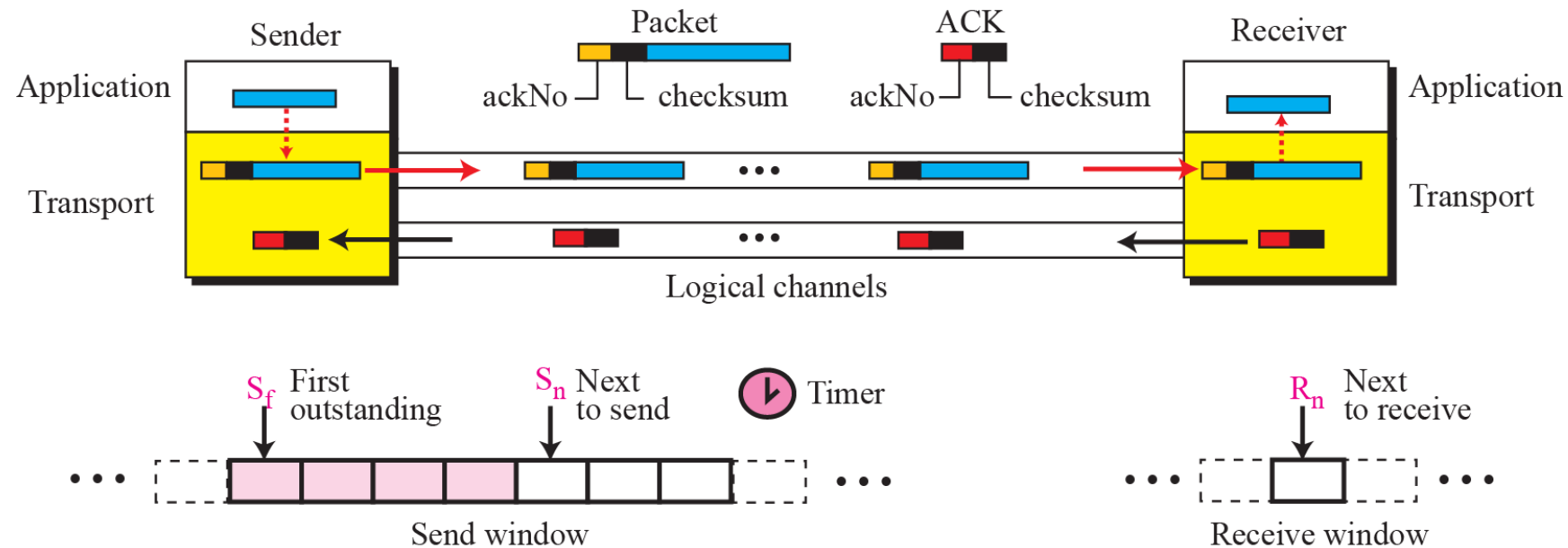


Transport Layer



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Go-Back-N protocol

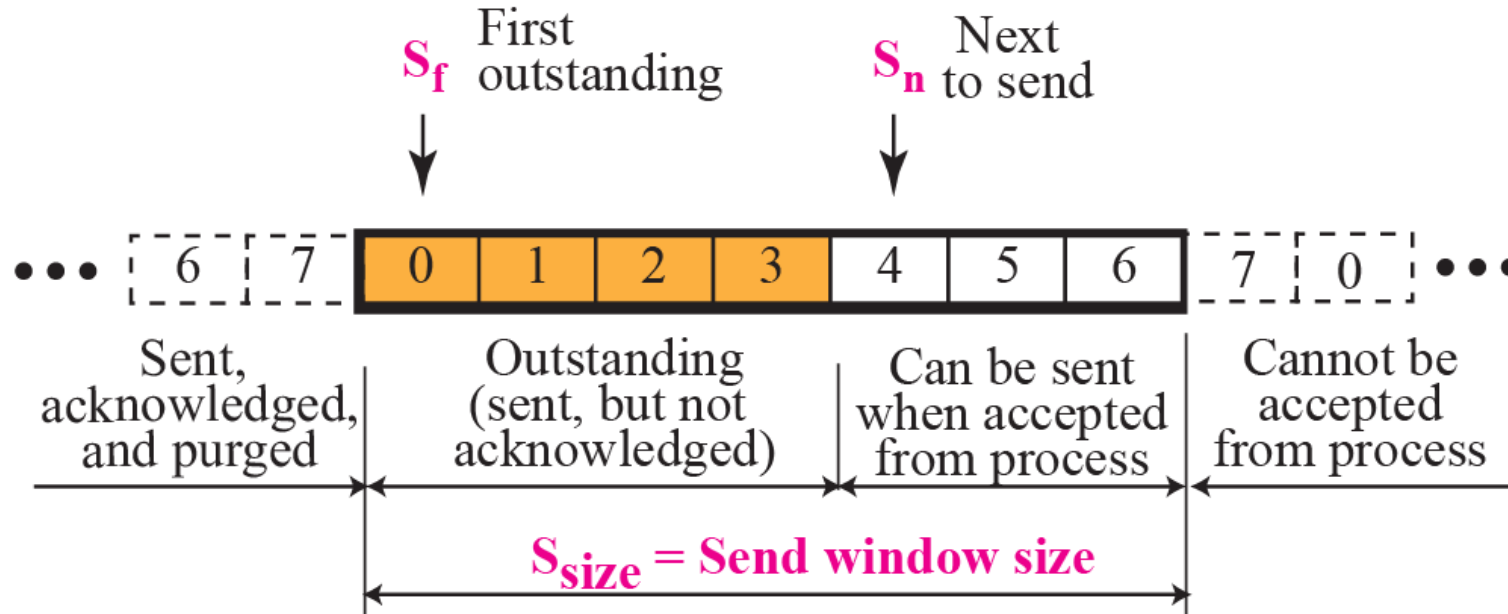


When the sender doesn't receive ACK, it retransmits the packet in error plus all the succeeding packets. Hence, the name of the protocol is go-back-N ARQ.

Go-Back-N protocol Cont..

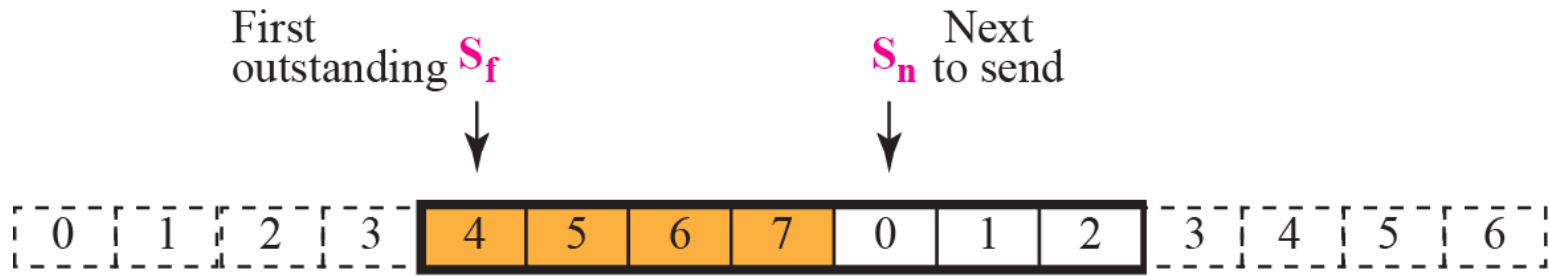
- In the Go-Back-N Protocol, the sequence numbers are modulo 2^m , where m is the size of the sequence number field in bits.
- In the Go-Back-N protocol, the acknowledgment number is cumulative and defines the sequence number of the next packet expected to arrive.

Send window for Go-Back-N

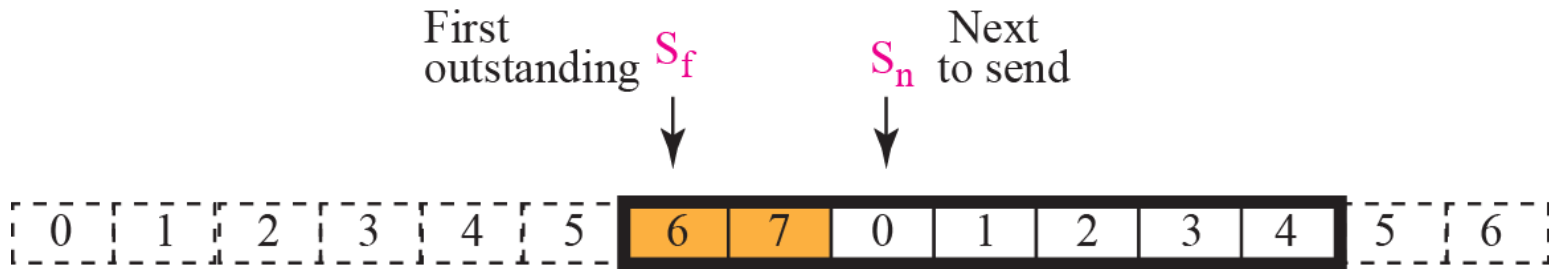


- The send window is an abstract concept defining an imaginary box of maximum size $= 2^m - 1$ with three variables: S_f , S_n , and S_{size} .
- The send window can slide one or more slots when an error-free ACK with ack No. between S_f and S_n (in modular arithmetic) arrives.

Sliding the send window

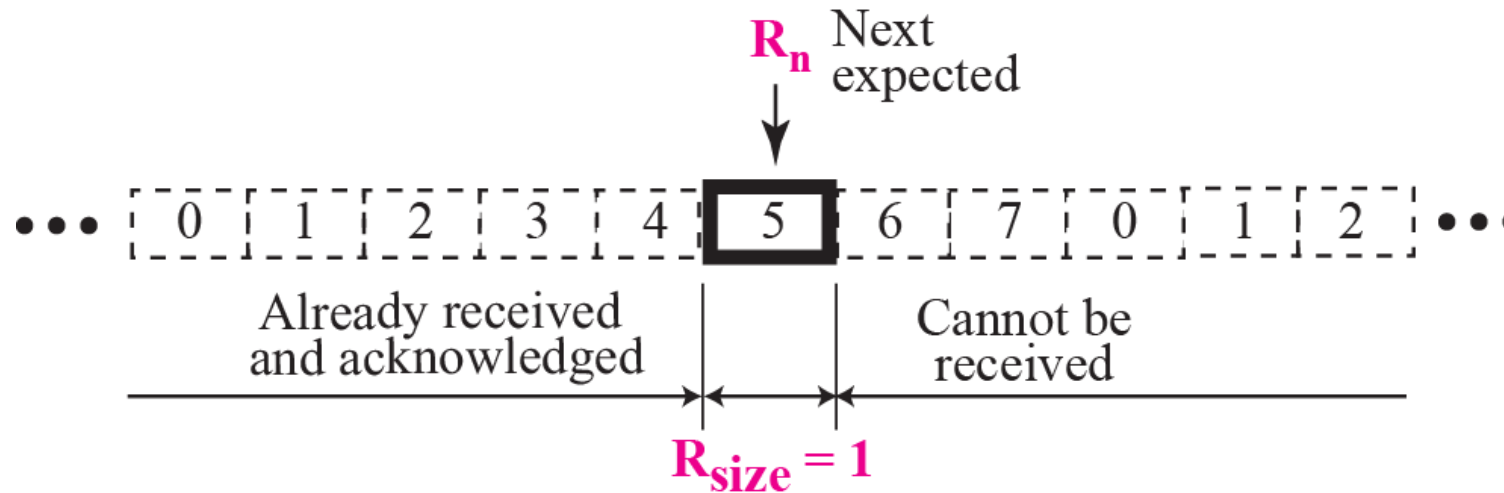


a. Window before sliding



b. Window after sliding (an ACK with ackNo = 6 has arrived)

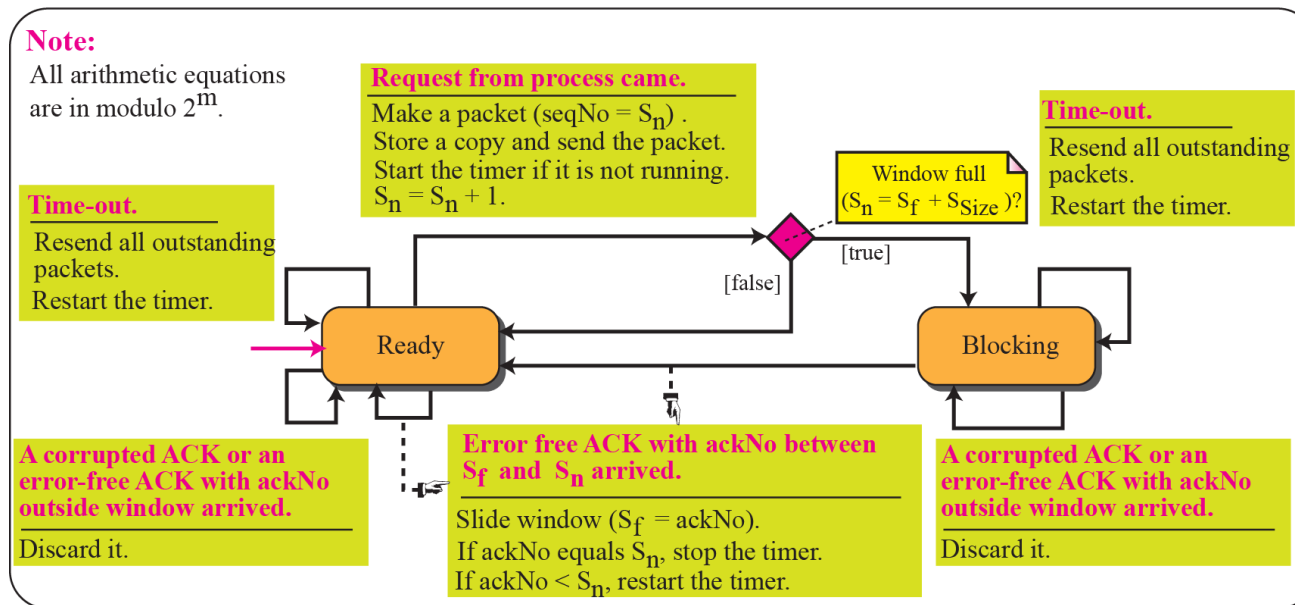
Receive window for Go-Back-N



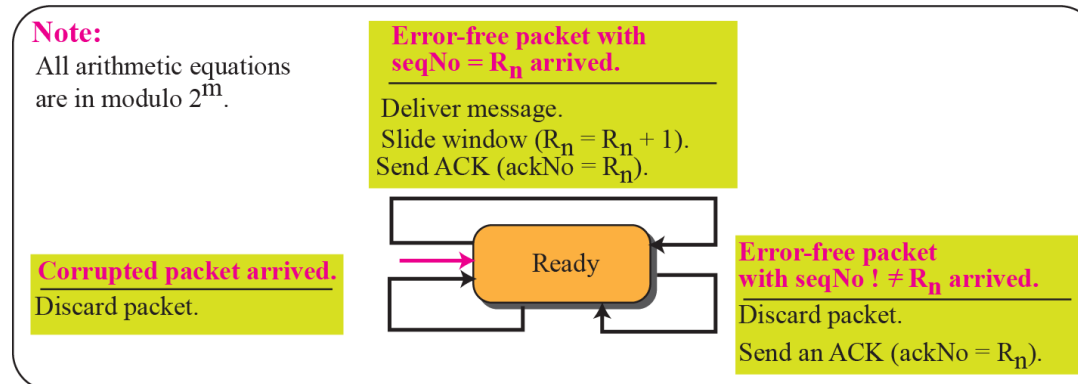
- The receive window is an abstract concept defining an imaginary box of size 1 with one single variable R_n .
- The window slides when a correct packet has arrived; sliding occurs one slot at a time.
- *In the Go-Back-N protocol, the size of the send window must be less than 2^m ; the size of the receive window is always 1.*

FSMs for Go-Back-N

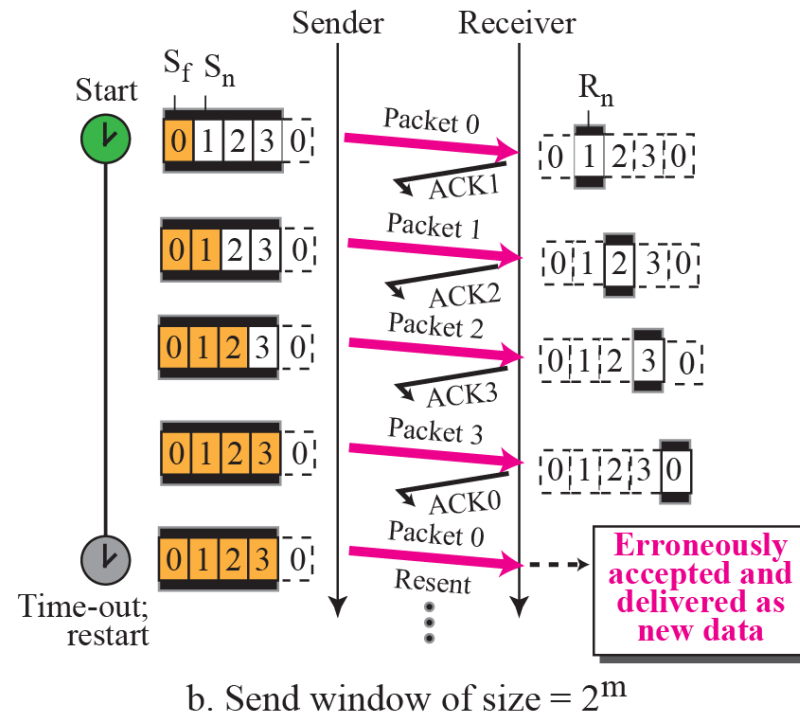
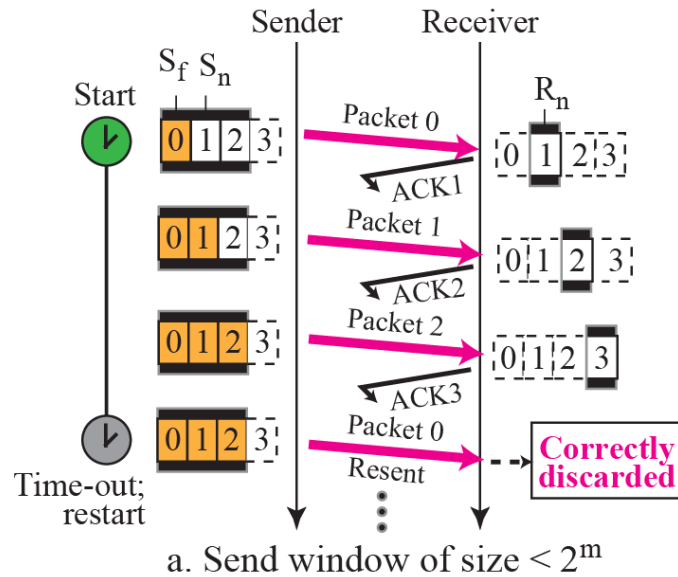
Sender



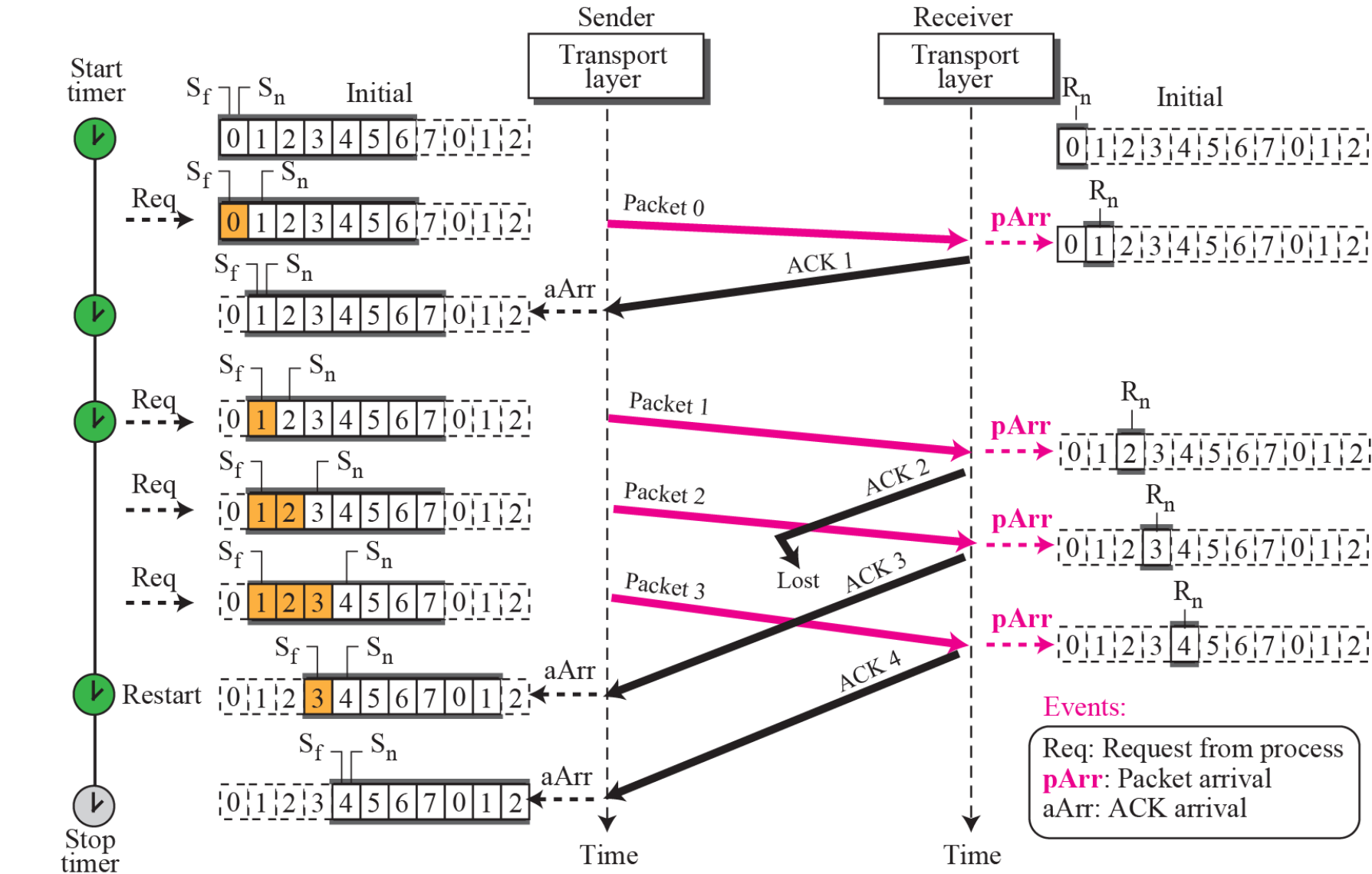
Receiver



Send window size for Go-Back-N

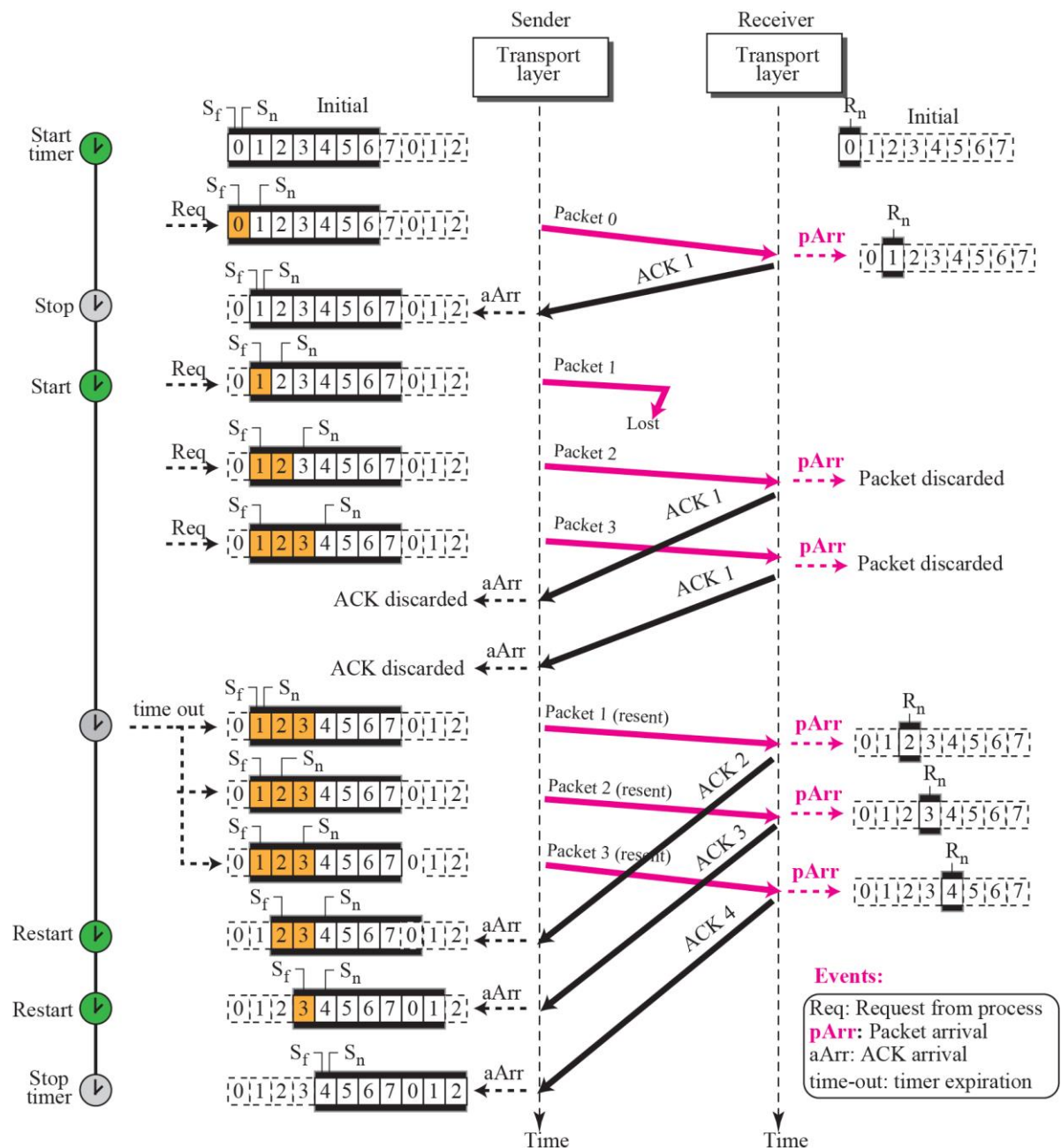


Example

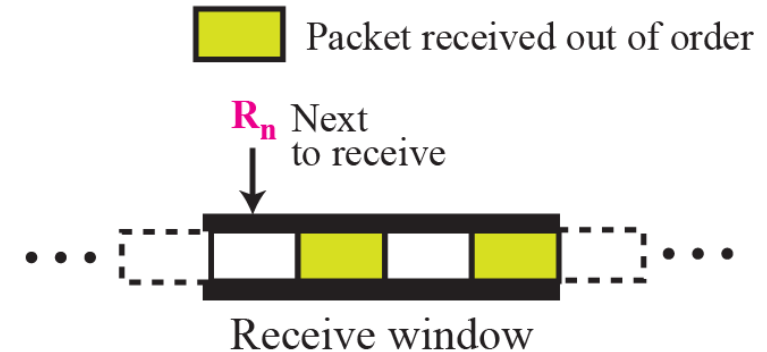
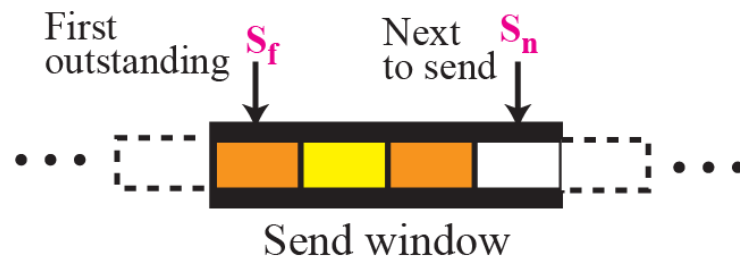
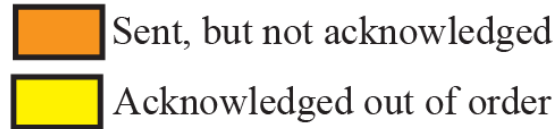
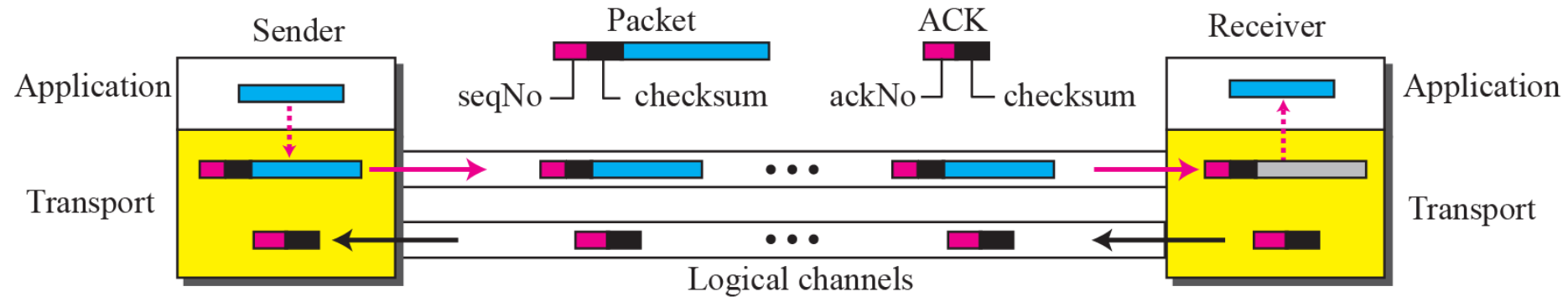


Cumulative
acknowledgments
can help

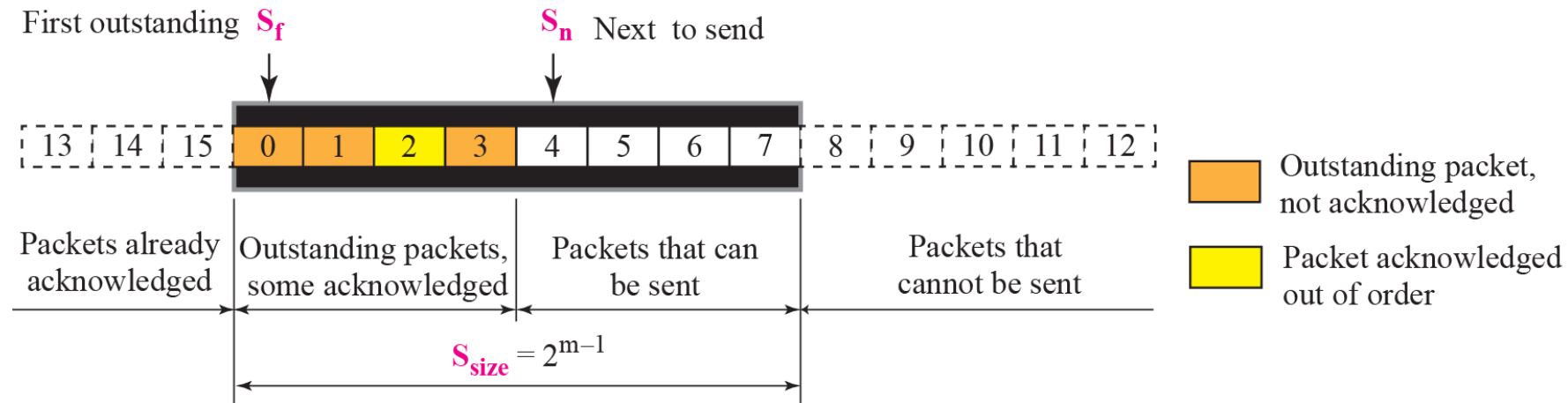
Example 2



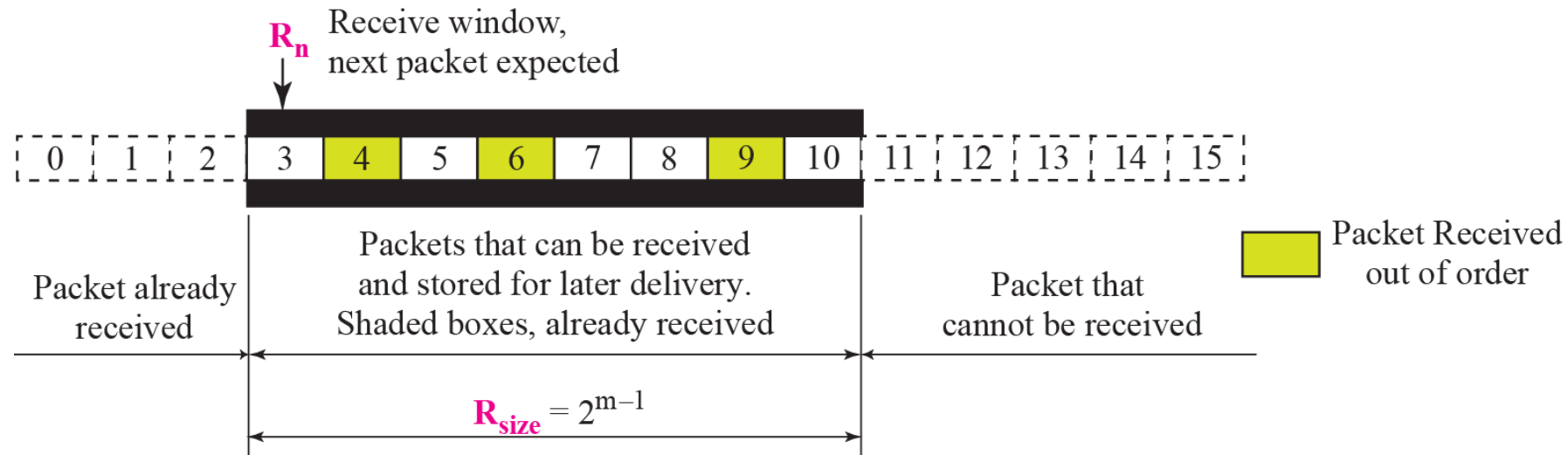
Outline of Selective-Repeat



Send window for Selective-Repeat protocol



Receive window for Selective-Repeat protocol

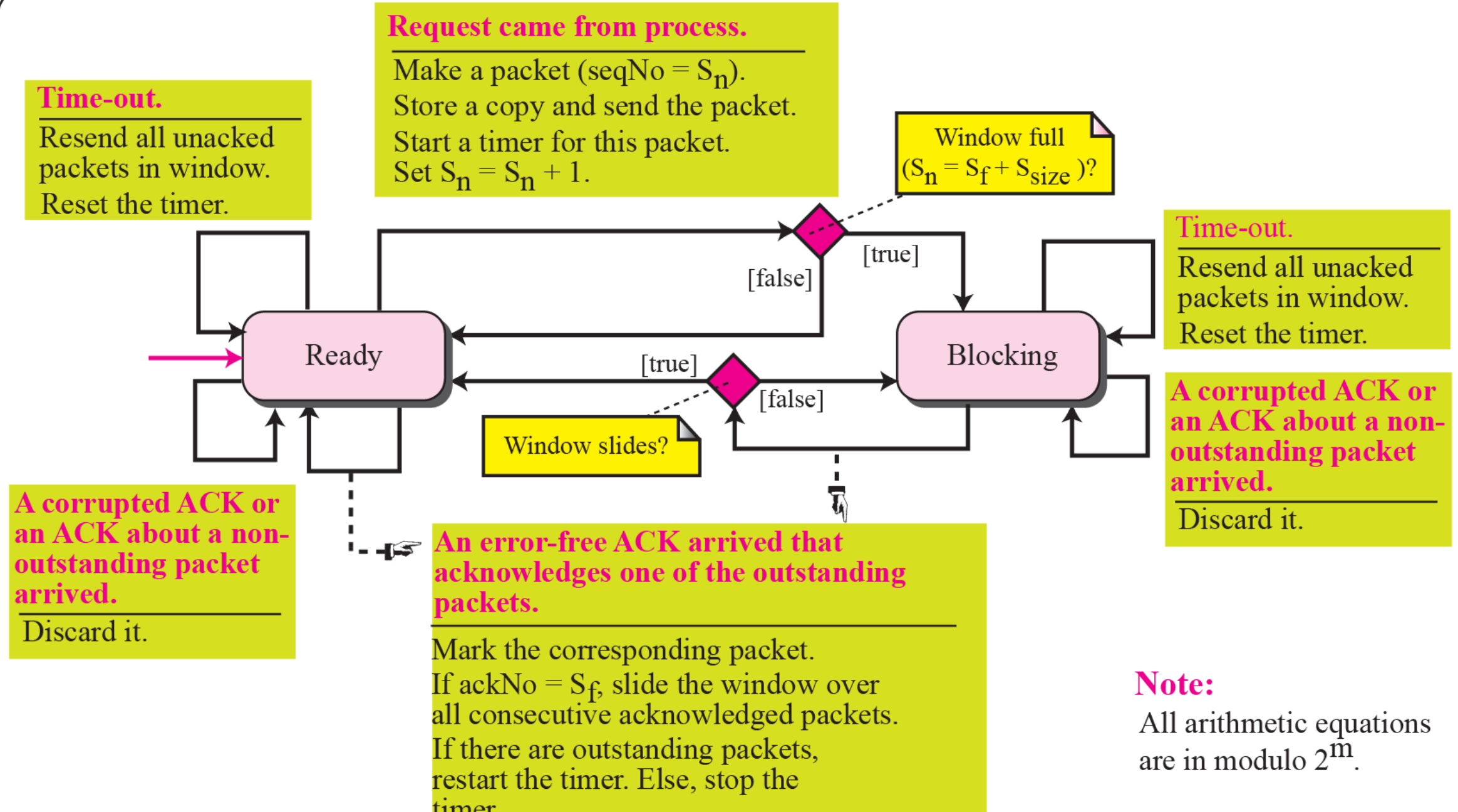


Example

Assume a sender sends 6 packets: packets 0, 1, 2, 3, 4, and 5. The sender receives an ACK with `ackNo = 3`. What is the interpretation if the system is using GBN or SR?

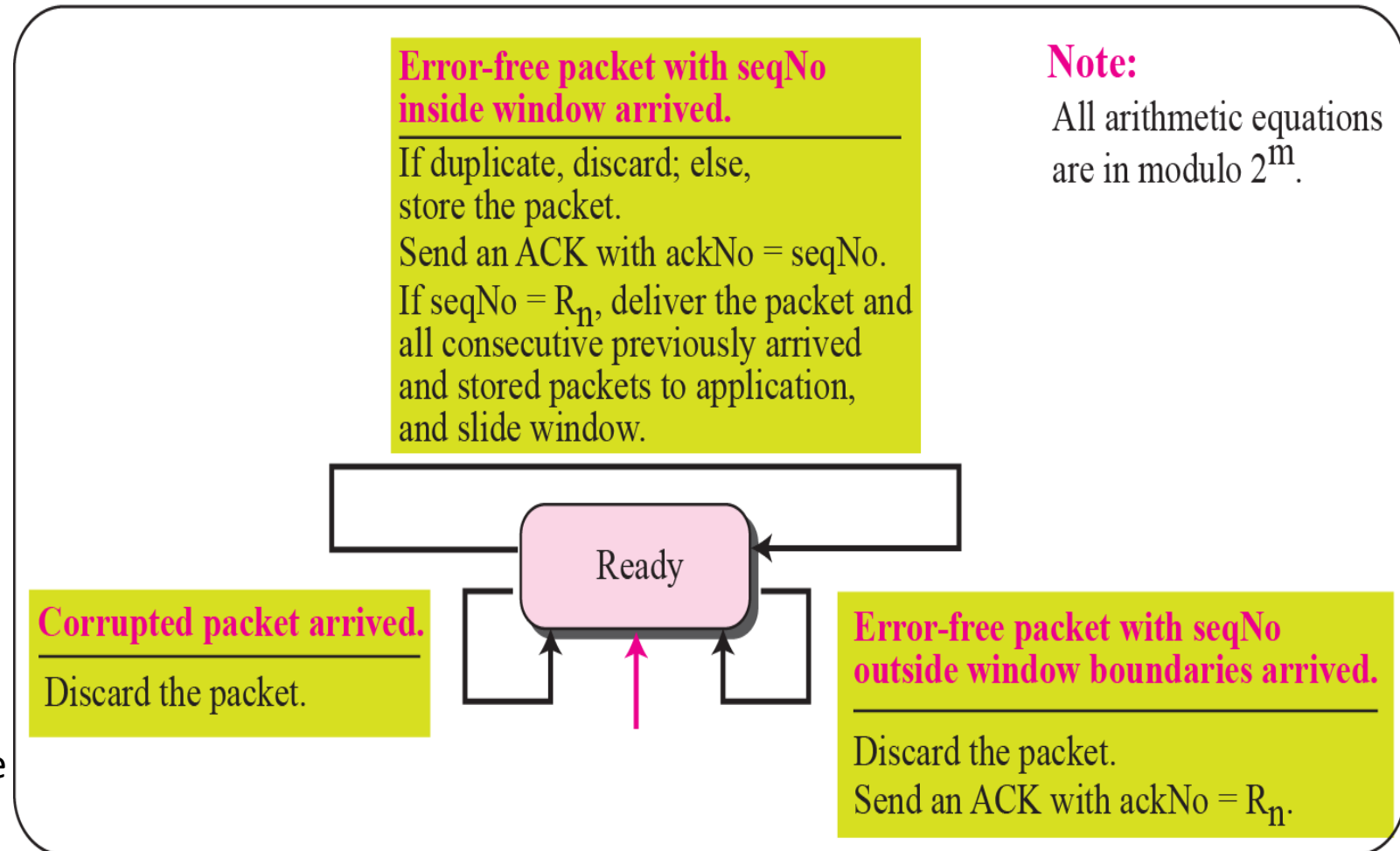
Solution

If the system is using GBN, it means that packets 0, 1, and 2 have been received uncorrupted and the receiver is expecting packet 3. If the system is using SR, it means that packet 3 has been received uncorrupted; the ACK does not say anything about other packets.



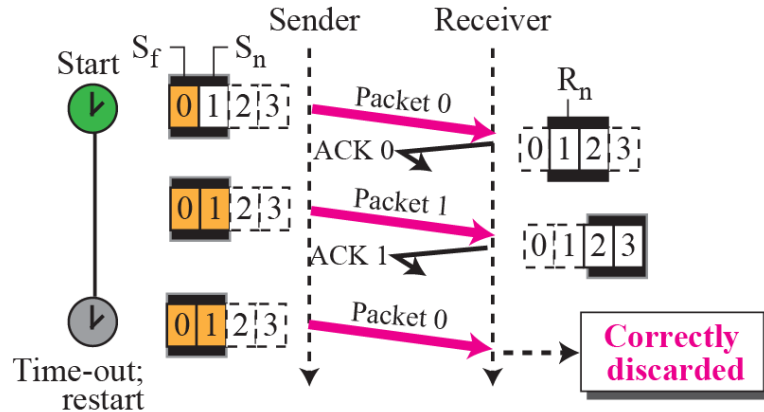
FSMs for SR protocol

Receiver

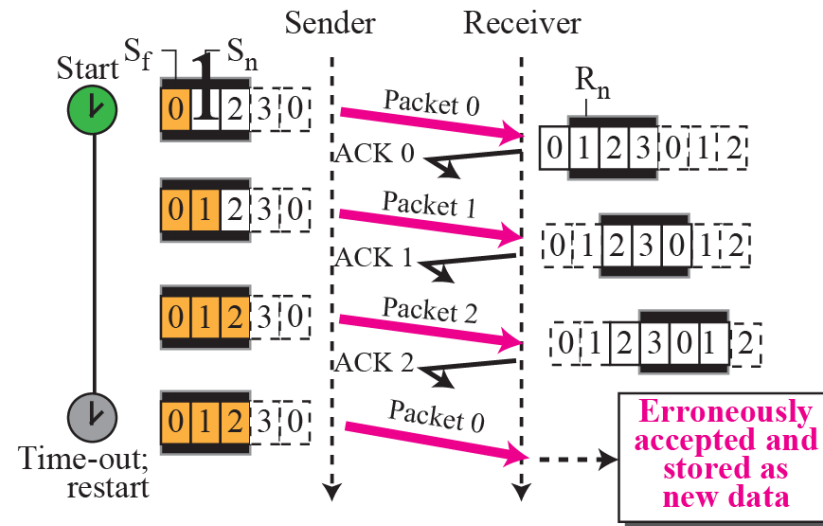


TCP/IP Protocol Suite

Selective-Repeat window size

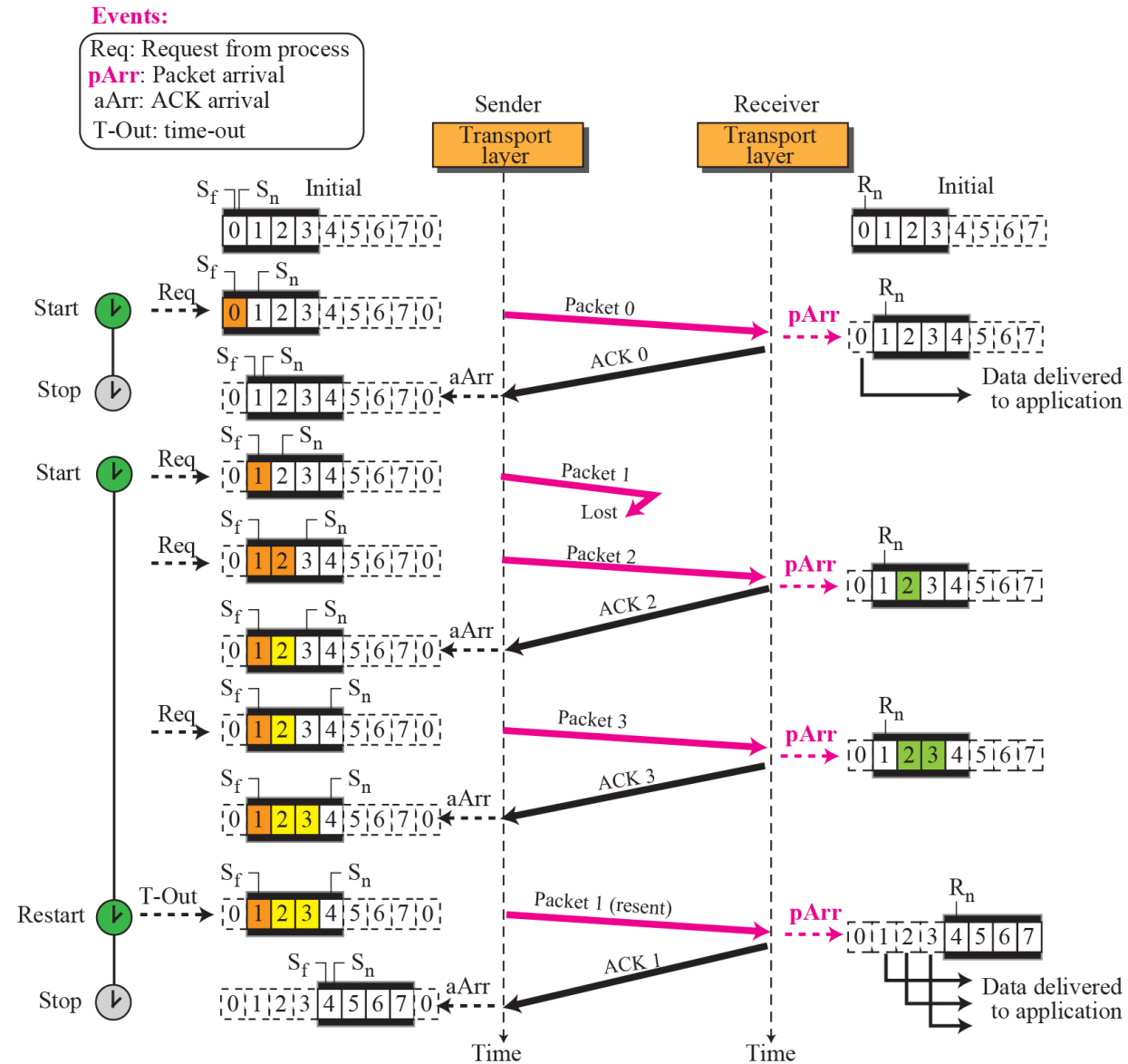


a. Send and receive windows of size $= 2^m - 1$



b. Send and receive windows of size $> 2^m - 1$

Example



Selective-Repeat

This is the most efficient among the ARQ schemes, but the sender must be more complex so that it can send out-of-order frames. The receiver also must have storage space to store the post-NAK frames and processing power to reinsert frames in proper sequence.